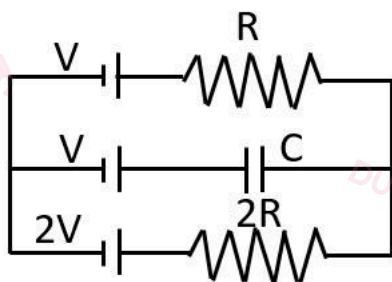


Q.1

In the given circuit when steady current flows, the potential drop across the capacitor is:



1. V
2. $V/3$
3. $V/2$
4. $2V/3$

Options 1. 1

2. 2
3. 3
4. 4

Q.2 The rate of change of current which produces an induced emf of 100 V in an inductor of inductance 10 H is

1. 8 A/s
2. 10 A/s
3. 12.5 A/s
4. 14 A/s

Options 1. 1

2. 2
3. 3
4. 4

Q.3 In a diffraction experiment, the screen is placed $D = 2$ m away from a narrow slit. The first minimum observed is at $x = 5$ mm on either side of the central maximum for wavelength of $\lambda = 5 \times 10^{-5}$ cm. The width of the slit a is:

1. 0.01 cm
2. 0.02 cm
3. 0.11 cm
4. 0.12 cm

Options 1. 1

2. 2
3. 3
4. 4

Q.4 A thin prism of refractive index 1.5 when placed in air deviates a light ray by a minimum angle of 5° . The angle of minimum deviation when it is immersed in oil of refractive index 1.25 is:

1. 2°
2. 4°
3. 5°
4. 2.5°

Options 1. 1

2. 2
3. 3
4. 4

Q.5 If the mass of a proton is $m_p = 1.007825$ u and the mass of the neutron is $m_n = 1.008665$ u, the mass defect in the formation of ${}^7_3\text{Li}$ nucleus is: (take the mass of Lithium nucleus as $M_{\text{Li}} = 7.000000$ u)

1. 0.058135 u
2. 0.057295 u
3. 0.016490 u
4. 0.056420 u

Options 1. 1

2. 2
3. 3
4. 4

Q.6 Arrange the following nuclei in the ascending order of their stability.

- (A) Oxygen (${}^{16}\text{O}$) with binding energy 127.5 MeV
- (B) Uranium (${}^{238}\text{U}$) with binding energy 1808.8 MeV
- (C) Helium (${}^4\text{He}$) with binding energy 28.3 MeV
- (D) Iron (${}^{56}\text{Fe}$) with binding energy 490 MeV

Choose the correct answer from the options given below:

1. (C), (D), (A), (B)
2. (C), (A), (D), (B)
3. (C), (B), (A), (D)
4. (C), (A), (B), (D)

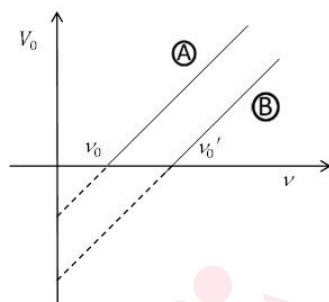
Options 1. 1

2. 2
3. 3
4. 4

Q.7

The graphs of stopping potential versus frequency of incident radiation for two metals (A) and (B) are given.

Identify the correct statement(s) from the following



- (A) The slope of the graphs is equal to the Planck's constant.
- (B) Intercept on the (-y) axis is equal to (work function/e).
- (C) Threshold frequency for metal (A) is higher than metal (B)
- (D) Work function for metal (B) is greater than metal (A)

Choose the **correct** answer from the options given below:

1. (D) only
2. (A) and (D) only
3. (A) and (C) only
4. (B) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.8 A conducting circular loop is placed in a uniform magnetic field of $B = 0.5$ T, with its plane perpendicular to the magnetic field. The radius of the loop starts shrinking at the rate of $\frac{dr}{dt} = 4$ mm/s. When the radius of the loop is $r = 4$ cm, the induced

emf in the loop is

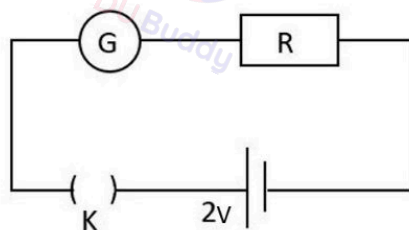
1. $3.2\pi \mu\text{V}$
2. $1.6\pi \mu\text{V}$
3. $0.8\pi \mu\text{V}$
4. $160\pi \mu\text{V}$

Options 1. 1

2. 2
3. 3
4. 4

Q.9

In the given circuit, the resistance of the galvanometer is $G = 100 \Omega$. Resistance box R contributes 4900Ω . When key K is closed. The needle of the galvanometer deflects to the 20th division. The figure of merit of the galvanometer is:



1. 2×10^{-5} A/div
2. 2×10^{-4} A/div
3. 5×10^{-5} A/div
4. 5×10^{-4} A/div

Options 1. 1

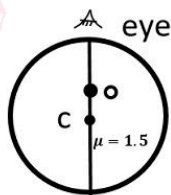
2. 2

3. 3

4. 4

Q.10

There is a small air bubble inside a glass sphere ($\mu = 1.5$) of radius 10 cm. The bubble is 4.0 cm below the surface and is viewed normally from outside as shown. The apparent depth of the bubble will be



1. 6.0 cm below the surface

2. 4.0 cm below the surface

3. 3.0 cm below the surface

4. 2.0 cm below the surface

Options 1. 1

2. 2

3. 3

4. 4

Q.11

A proton moves with a velocity equal to $\left(\frac{1}{20}\right)^{\text{th}}$ of velocity of light.

The associated de-Broglie wavelength is:

1. $1.3 \times 10^{-12} \text{ m}$

2. $1.98 \times 10^{-14} \text{ m}$

3. $2.65 \times 10^{-14} \text{ m}$

4. $1.6 \times 10^{-12} \text{ m}$

Options 1. 1

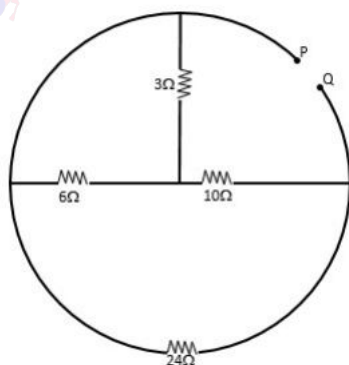
2. 2

3. 3

4. 4

Q.12

The equivalent resistance between P and Q is:



1. 8Ω

2. 7.5Ω

3. 16Ω

4. 12Ω

Options 1. 1

2. 2

3. 3

4. 4

Q.13 Two conducting spheres A and B of radii 'a' and 'b', respectively are charged to have the same electric potential. The ratio of surface charge densities of A and B is

1. $\frac{a}{b}$
2. $\frac{b}{a}$
3. $\frac{a^2}{b^2}$
4. $\frac{b^2}{a^2}$

Options 1. 1
2. 2
3. 3
4. 4

Q.14 A cylindrical wire is stretched to increase its length by 10%. The percentage increase in resistance will be

1. 10%
2. 15%
3. 21%
4. 32%

Options 1. 1
2. 2
3. 3
4. 4

Q.15 Match List-I with List-II

List-I

List-II

- | | |
|------------------------------------|--|
| (A) Intrinsic semiconductor | (I) Number of conduction electrons >> number of holes |
| (B) n-type semiconductor | (II) Number of conduction electrons << number of holes |
| (C) p-type semiconductor | (III) Net current is zero |
| (D) p-n junction under equilibrium | (IV) Number of conduction electrons = number of holes |

Choose the correct answer from the options given below:

1. (A) - (II), (B) - (IV), (C) - (I), (D) - (III)
2. (A) - (IV), (B) - (II), (C) - (I), (D) - (III)
3. (A) - (IV), (B) - (I), (C) - (II), (D) - (III)
4. (A) - (II), (B) - (I), (C) - (IV), (D) - (III)

Options 1. 1
2. 2
3. 3
4. 4

Q.16 Which of the following has zero average value in a plane electromagnetic wave?

- (A) Electric field
- (B) Magnetic field
- (C) Electric energy
- (D) Magnetic energy

Choose the correct answer from the options given below:

1. (A) and (C) only
2. (B) and (D) only
3. (B) only
4. (A) and (B) only

Options 1. 1
2. 2
3. 3
4. 4

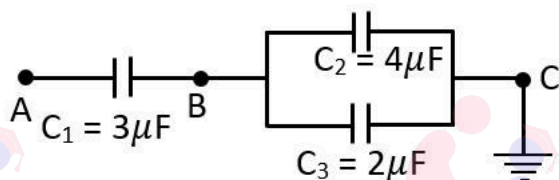
Q.17 When light passes from air to glass, then

1. its wavelength increases and velocity decreases.
2. its wavelength and velocity both decrease.
3. its wavelength decreases but velocity increases.
4. its wavelength and velocity both increase.

Options 1. 1
2. 2
3. 3
4. 4

Q.18

In the given circuit, if the potential at A is 50 V, then potential at B is:



1. $\frac{50}{3}$ V
2. $\frac{100}{3}$ V
3. 50 V
4. 25 V

Options 1. 1
2. 2
3. 3
4. 4

Q.19 A piece of copper and a piece of germanium are cooled from room temperature 20°C down to 80 K. The resistance of

1. both of them will increase.
2. both of them will decrease.
3. copper increases and that of germanium decreases.
4. copper decreases and that of germanium increases.

Options 1. 1
2. 2
3. 3
4. 4

Q.20 An external voltage is applied across a diode such that the n-side is positive and the p-side is negative. As a result,

- (A) the barrier height decreases
- (B) the barrier height increases
- (C) the depletion region widens
- (D) the depletion region width decreases
- (E) no change in barrier height

Choose the correct answer from the options given below:

1. (A) and (B) only
2. (C) and (E) only
3. (D) and (E) only
4. (B) and (C) only

Options 1. 1
2. 2
3. 3
4. 4

Q.21 Rainbow is a phenomenon due to the combined effect of

- (A) Interference
- (B) Dispersion
- (C) Reflection
- (D) Refraction
- (E) Scattering

Choose the correct answer from the options given below:

1. (A), (B) and (C) only
2. (B), (C) and (D) only
3. (A) only
4. (C), (D) and (E) only

Options 1. 1
2. 2
3. 3
4. 4

Q.22 Two identical metal balls, A and B, with similar charges Q are suspended with insulating threads. They repel each other with a force F . Another identical uncharged metal ball C is first touched with metal ball A and then touched with metal ball B and finally removed. The new force of repulsion between A and B will be

1. F
2. $2F$
3. $\frac{1}{8}F$
4. $\frac{3}{8}F$

Options 1. 1
2. 2
3. 3
4. 4

Q.23 The ratio of radii of the nuclei with the mass numbers given as 2, 16, 54 and 128, respectively, will be

1. 1:3:4:8
2. 1:2:3:4
3. 2:3:4:5
4. 1:3:5:7

Options

1. 1
2. 2
3. 3
4. 4

Q.24 When a magnet is being brought towards a coil, the emf induced in the coil does not depend on:

1. the number of turns in the coil
2. the speed of the magnet
3. the pole strength of the magnet
4. the resistance of the coil

Options

1. 1
2. 2
3. 3
4. 4

Q.25 Voltage sensitivity of a moving coil galvanometer doesn't depend on :

1. Magnetic field
2. Number of turns of the coil
3. Resistance of the coil
4. torsional constant of the spring

Options

1. 1
2. 2
3. 3
4. 4

Q.26 Two cells of emf E and $2E$ with internal resistances r and $3r$, respectively are connected in parallel. The emf and internal resistance of the combination is:

1. $\frac{4}{3}E, \frac{2}{3}r$
2. $\frac{1}{4}E, \frac{3}{2}r$
3. $\frac{5}{4}E, \frac{3}{4}r$
4. $\frac{2}{3}E, \frac{1}{2}r$

Options

1. 1
2. 2
3. 3
4. 4

Q.27 An electron is projected into a magnetic field of flux density 10 Wb/m^2 with a velocity of $3 \times 10^7 \text{ m/s}$ at an angle of 30° to the field. The magnetic force on the electron is:

1. $1.6 \times 10^{-11} \text{ N}$
2. $4.8 \times 10^{-11} \text{ N}$
3. $2.4 \times 10^{-11} \text{ N}$
4. $9.2 \times 10^{-11} \text{ N}$

Options

1. 1
2. 2
3. 3
4. 4

Q.28 Identify the correct statement(s).

- (A) In any lens, the optical centre is always at the centre of the lens.
- (B) In a lens, if the curvature of its two surfaces is equal, the optical centre is at the centre of the lens.
- (C) A light ray passing through the optical centre of a lens never shifts from the direction of the incident ray.
- (D) A ray of light incident perpendicularly on a lens passes through its focus after refraction.

Choose the correct answer from the following given options.

1. (A) and (B) only
2. (B) only
3. (A), (B), (C) and (D)
4. (B) and (C) only

Options

1. 1
2. 2
3. 3
4. 4

Q.29 Kirchhoff's second law for an electric network is based on:

1. Law of conservation of mass
2. Law of conservation of charge
3. Law of conservation of energy
4. Law of conservation of momentum

Options

1. 1
2. 2
3. 3
4. 4

Q.30 An electron in a hydrogen atom emits wavelengths during the transitions given below. Arrange these wavelengths in ascending order.

- (A) from $n = 3$ to $n = 2$
- (B) from $n = 2$ to $n = 1$
- (C) from $n = \infty$ to $n = 1$
- (D) from $n = \infty$ to $n = 3$
- (E) from $n = 4$ to $n = 1$

Choose the correct answer from the options given below:

1. (C), (B), (E), (A), (D)
2. (C), (E), (B), (D), (A)

3. (C), (B), (A), (E), (D)

4. (C), (E), (B), (A), (D)

Options 1. 1

2. 2

3. 3

4. 4

Q.31 Match List-I with List-II

List-I

List-II

Phenomenon Characteristics

(A) Reflection (I) Used for reducing glare

(B) Refraction (II) Change in path when a light is incident on opaque medium

(C) Interference (III) Change in path when a light is incident on transparent medium

(D) Polarisation (IV) Light added to light to produce darkness

Choose the correct answer from the options given below:

1. (A) - (III), (B) - (I), (C) - (IV), (D) - (II)

2. (A) - (II), (B) - (III), (C) - (I), (D) - (IV)

3. (A) - (III), (B) - (II), (C) - (IV), (D) - (I)

4. (A) - (II), (B) - (III), (C) - (IV), (D) - (I)

Options 1. 1

2. 2

3. 3

4. 4

Q.32 When radiation is incident on a metal surface of work function Φ_0 , photoelectrons are emitted with some maximum kinetic energy K . If the frequency of the incident radiation is doubled, then the maximum kinetic energy of the emitted photoelectrons will become

1. $2K$

2. $2K - \Phi_0$

3. $2K + \Phi_0$

4. K

Options 1. 1

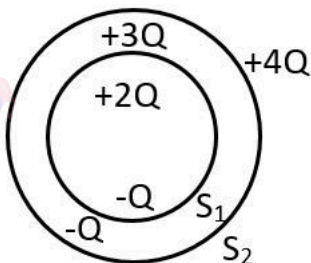
2. 2

3. 3

4. 4

Q.33

In the given figure, the ratio of electric fluxes through surfaces S_1 and S_2 is



1. 1 : 3

2. 1 : 6

3. 1 : 7

4. 1 : 2

Options 1. 1

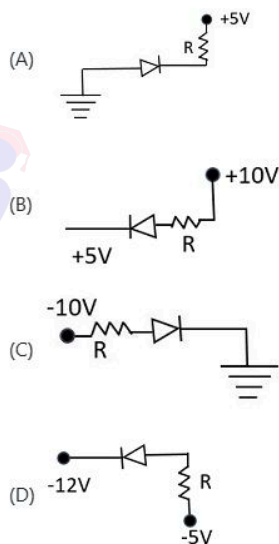
2. 2

3. 3

4. 4

Q.34

Four diode connections are given below. Identify the diagram in which the diode is forward biased.



Choose the **correct** answer from the options given below:

1. (B) only
2. (B) and (D) only
3. (B), (C) and (D) only
4. (B) and (C) only

Options

1. 1
2. 2
3. 3
4. 4

Q.35 Identify the electromagnetic waves as per their following use/ applications.

- (A) λ_1 is used to produce heating effect
 (B) λ_2 is used in satellite communication
 (C) λ_3 is absorbed by the ozone layer
 (D) λ_4 is used for studying crystal structure

Choose the correct answer from the options given below:

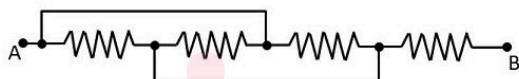
1. (A) Microwave, (B) Visible, (C) UV, (D) Infrared
2. (A) Infrared, (B) Microwave, (C) Ultraviolet, (D) Visible
3. (A) Infrared, (B) Microwave, (C) Ultraviolet, (D) X-rays
4. (A) UV, (B) Microwave, (C) Infrared, (D) Visible

Options

1. 1
2. 2
3. 3
4. 4

Q.36

Four resistors of $6\ \Omega$ resistance each are joined as shown in the figure. The equivalent resistance of the combination is:



1. $8\ \Omega$
2. $10.5\ \Omega$
3. $24\ \Omega$
4. $1.5\ \Omega$

Options

1. 1
2. 2
3. 3
4. 4

Q.37 A step-up transformer operates on a 220 V line and supplies a current of 2 A. The ratio of primary and secondary winding is 1 : 25. Assuming 100% efficiency, the output power of the transformer is:

1. 11000 W
2. 22000 W
3. 36000 W
4. 44000 W

Options

1. 1
2. 2
3. 3

Q.38 The magnetic field in a plane electromagnetic wave is given by $B = 3 \times 10^{-7} \sin(1.5x + 5 \times 10^8 t)$ T, where x is in meter and t is in second. The wavelength of the wave is: (Take $\pi = 3$)

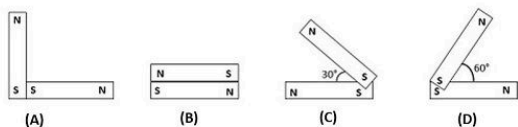
1. 1.5 m
2. 3 m
3. 4 m
4. 2.5 m

Options 1. 1

2. 2
3. 3
4. 4

Q.39

The figure shows an arrangement of bar magnets in different configurations. Each bar magnet has a magnetic dipole moment m . The figure which has the maximum magnetic dipole moment _____.



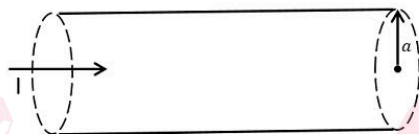
1. configuration (A)
2. configuration (B)
3. configuration (C)
4. configuration (D)

Options 1. 1

2. 2
3. 3
4. 4

Q.40

The figure shows a long cylindrical straight wire of radius a carrying steady current (I). The current (I) is uniformly distributed across the cross-section of the wire. The ratio of the magnetic fields at points $r = \frac{a}{3}$ and $r = \frac{2}{3}a$ respectively is : Here r is the distance of point from the axis of cylinder.



1. 2 : 3
2. 3 : 2
3. 9 : 4
4. 1 : 1

Options 1. 1

2. 2
3. 3
4. 4

Q.41 A transformer has 400 turns in the primary coil and 8000 turns in the secondary coil. It is connected to a 220 V main supply. The potential difference per turn in secondary coil is

1. 1.1 V
2. 11 V
3. 0.55 V
4. 5.5 V

Options 1. 1

2. 2
3. 3
4. 4

Q.42 The intensity of transmitted light is maximum when a Polaroid sheet (P_2) is rotated between two crossed Polaroids, P_1 and P_3 when

1. the angle between P_1 and P_2 and P_2 and P_3 will be $\frac{\pi}{4}$
2. Polaroids P_1 and P_2 are crossed
3. angle between P_1 and P_2 is $\frac{\pi}{3}$ and between P_2 and P_3 is $\frac{\pi}{6}$
4. the angle between P_2 and P_3 does not matter

Options 1. 1

2. 2
3. 3
4. 4

Q.43 The magnetic field strength in silicon is 1000 A/m. If its magnetic susceptibility is -0.3×10^{-5} , then the intensity of magnetization will be:

1. 1.3×10^{-3} A/m

2. $-3 \times 10^{-3} \text{ A/m}$
 3. $4.5 \times 10^{-3} \text{ A/m}$
 4. $-6.0 \times 10^{-3} \text{ A/m}$

Options 1. 1
 2. 2
 3. 3
 4. 4

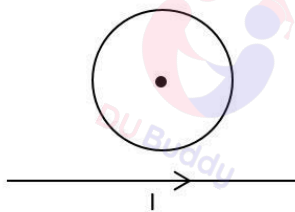
Q.44 An electric dipole is kept in an electric field which is increasing in the $+x$ direction. The dipole moment of the dipole is also along the $+x$ direction. Which of the following options is correct for the electric dipole?

1. Electric dipole will not experience any force in the presence of the electric field ($\vec{E}$).
2. Electric dipole will experience force and try to align perpendicular to the direction of electric field ($\vec{E}$).
3. Electric dipole will experience a net force in $+x$ direction.
4. Electric dipole will experience a net force in $-x$ direction.

Options 1. 1
 2. 2
 3. 3
 4. 4

Q.45

A circular loop of conducting wire is placed near a current carrying wire as shown in the figure. Both are in the same plane. If the current flowing through the wire is suddenly increased.



1. Loop will have induced current in the anti-clockwise direction
2. Loop will have induced current in the clockwise direction
3. There will be no induced current in the loop
4. the loop will be attracted by the wire

Options 1. 1
 2. 2
 3. 3
 4. 4

Q.46 An electron is moving at a velocity of $3.2 \times 10^7 \text{ m/s}$ in a magnetic field along a circular path of radius 20 cm. The magnitude of the magnetic field applied is (approximately)

1. $1.6 \times 10^{-4} \text{ T}$
2. $8.43 \times 10^{-5} \text{ T}$
3. $3.9 \times 10^{-4} \text{ T}$
4. $4.9 \times 10^{-5} \text{ T}$

Options 1. 1
 2. 2
 3. 3
 4. 4

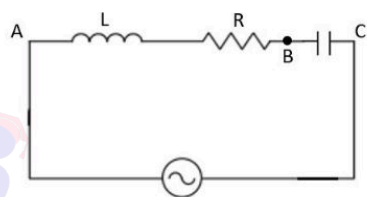
Q.47 Three point charges $+q, +q$ and $-2q$ are placed at the corner of an equilateral triangle of side 'a'. The net dipole moment of the system is

1. Zero
2. $2qa$
3. $\frac{1}{\sqrt{3}}qa$
4. $qa\sqrt{3}$

Options 1. 1
 2. 2
 3. 3
 4. 4

Q.48

An LCR series circuit, with $L = \frac{2}{\pi}$ H, $R = 100 \Omega$ and $C = \frac{100}{\pi} \mu\text{F}$ is connected across a source of 200 V, 50 Hz as shown. The peak potential difference between A and B is:



1. 600 V
2. $200\sqrt{5}$ V
3. $20\sqrt{5}$ V
4. $300\sqrt{2}$ V

- Options
1. 1
 2. 2
 3. 3
 4. 4

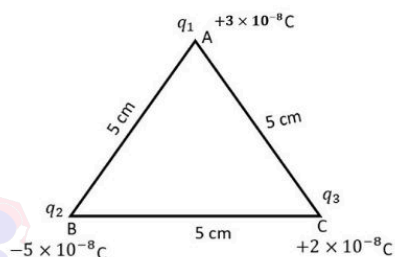
Q.49 Which of the following statements is correct for a charged metallic shell?

1. The electric field intensity is zero inside the shell and decreases outside the shell as we move away from the shell.
2. The electric field intensity is uniform inside the shell and decreases outside the shell as we move away from its center.
3. The electric potential is zero inside the shell and decreases outside the shell as we move away from its center.
4. The electrostatic potential increases linearly with the distance from its center inside the shell.

- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.50

Three point charges are placed at the corners of an equilateral triangle ABC as shown. The work done to put together these charges from infinity to their respective locations is



1. 3.42×10^{-4} J
2. 34.2×10^{-4} J
3. -3.42×10^{-4} J
4. -5.5×10^{-4} J

- Options
1. 1
 2. 2
 3. 3
 4. 4